

INDUSTRY PROBLEM-SOLVING TASK

Java AWT, Applets, Event Handling and Layout Managers

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Programme / Section	Java	Date: 23-08-2026	
Course Code	CSA09	Course: Java Programming	Programming in Java
Faculty	Dr. C. Sivashankar	Duration:	Lab Session

INDUSTRY TASK 1: Employee Performance & Payroll Dashboard

Problem Statement: A software company wants to develop a simple Employee Performance and Payroll Management System using Java AWT. The application should allow the HR team to enter employee details, process salary information, and generate a structured payroll report.

Functional Requirements

- Develop an AWT-based graphical user interface.
- Accept Employee ID, Employee Name, Department, Designation, Basic Salary, Working Days, and Performance Rating.
- Use suitable AWT components such as Label, TextField, Choice, Checkbox/CheckboxGroup, Button, TextArea, and List.
- Calculate Gross Salary, Performance Bonus, and Net Salary using suitable business rules defined by the student.
- Display the complete employee payroll report in a List or TextArea when Generate Report is clicked.
- Use GridBagLayout or BorderLayout for arranging the interface.
- Implement button events using ActionListener.
- Provide Submit/Generate Report and Reset functionality.
- Validate important inputs such as empty fields and numeric salary values.

Industry Context: The application simulates an HR payroll module used to capture employee information, perform basic payroll processing, and present a report for HR verification.

Expected Deliverables

- Java source code.
- Working GUI application with event handling.
- Sample input and output.
- Screenshot of the final application.
- Brief explanation of components, layout manager, and event handling used.

JAVA PROGRAMMING LABORATORY

Code:

```
File Edit View

import java.awt.*;
import java.awt.event.*;

public class TASK1 extends Frame implements ActionListener {

    Label lblId, lblName, lblDept, lblDesig;
    Label lblBasic, lblDays, lblRating;

    TextField txtId, txtName, txtBasic, txtDays;

    Choice deptChoice, desigChoice;

    CheckboxGroup ratingGroup;
    Checkbox r1, r2, r3, r4, r5;

    Button btnGenerate, btnReset;

    TextArea reportArea;

    List employeeList;

    public TASK1() {

        setTitle("Employee Performance & Payroll Dashboard");
        setSize(850, 650);
        setLayout(new GridBagLayout());
```

Output:

The screenshot shows a Windows desktop environment. On the left, a Windows Command Prompt window displays the following commands and output:

```
C:\WINDOWS\system32\cmd
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Asmitha kadi\Downloads>cd downloads
C:\Users\Asmitha kadi\Downloads>java -version
openjdk version "21.0.12.1" 2026-08-18 LTS
OpenJDK Runtime Environment Temurin-21.0.12.1+1 (build 21.0.12.1+1-LTS)
OpenJDK 64-Bit Server VM Temurin-21.0.12.1+1 (build 21.0.12.1+1-LTS, mixed mode, sharing)

C:\Users\Asmitha kadi\Downloads>set path=C:\Program Files\Java\jdk1.8.0_202\bin
C:\Users\Asmitha kadi\Downloads>javac -version
javac 1.8.0_202

C:\Users\Asmitha kadi\Downloads>javac TASK1.java
C:\Users\Asmitha kadi\Downloads>java TASK1
```

On the right, the 'Employee Performance & Payroll Dashboard' application window is shown. It contains a form with the following fields:

- Employee ID:
- Employee Name:
- Department:
- Designation:
- Basic Salary:
- Working Days:
- Performance Rating: ☐ 1 ☐ 2 ☐ 3 ☒ 4 ☐ 5

Below the form are two buttons: 'Generate Report' and 'Reset'. The 'Generate Report' button is active, and a small window displays the following payroll details:

Payroll Reports	
Employee Performance	
Salary Details	
Basic Salary	: 69230.77
HRA (20%)	: 13846.15
DA (10%)	: 6923.08
Gross Salary	: 90000.00
Performance Bonus	: 13500.00
PF (12%)	: 8307.69
NET SALARY	: 95192.31

INDUSTRY TASK 2: Online Food Ordering and Billing System

Problem Statement: A restaurant wants to develop a simple Food Ordering and Billing Application for its counter staff. The application should allow customers to select food items, specify quantities, and generate a final bill.

Functional Requirements

- Develop the application using Java AWT.
- Create fields for Customer Name and Table Number.
- Provide Food Category, Food Item, and Quantity selection using appropriate AWT components.
- Provide Add Item, Generate Bill, Clear, Next, and Previous buttons.
- Display selected food items in a List or TextArea.
- Calculate item-wise cost, Subtotal, GST, and Final Bill Amount.
- Implement button click events using ActionListener.
- Use CardLayout to create separate screens for Customer/Order Details, Food Selection, and Bill/Order Summary.
- Validate quantity and ensure that an item is selected before adding it to the order.

Industry Context: The application represents a simplified restaurant POS (Point of Sale) system used for taking customer orders, processing prices, and generating a customer bill.

Expected Deliverables

- Java source code.
- Working multi-card AWT interface.
- Sample order and generated bill.
- Screenshot of each card/screen.
- Brief explanation of CardLayout, AWT components, and event handling.

Code:

JAVA PROGRAMMING LABORATORY

```
import java.awt.*;
import java.awt.event.*;

public class TASK2 extends Frame implements ActionListener {

    CardLayout cardLayout;
    Panel cardPanel;

    // Customer details
    TextField txtCustomerName;
    TextField txtTableNumber;

    // Food selection
    Choice categoryChoice;
    Choice itemChoice;
    TextField txtQuantity;

    TextArea orderArea;

    // Bill
    TextArea billArea;

    Button nextButton1;
    Button previousButton2;
    Button nextButton2;
}
```

Output:

The screenshot displays a Windows command prompt window on the left and an application window titled "Online Food Ordering and Billing System" on the right.

Command Prompt Window:

```
C:\WINDOWS\system32\CMD
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Asmitha kadiri>cd downloads

C:\Users\Asmitha kadiri\Downloads>java -version
openjdk version "21.0.12.1" 2026-08-18 LTS
OpenJDK Runtime Environment Temurin-21.0.12.1+1 (build 21.0.12.1+1-LTS)
OpenJDK 64-Bit Server VM Temurin-21.0.12.1+1 (build 21.0.12.1+1-LTS, mixed mode, sharing)

C:\Users\Asmitha kadiri\Downloads>set path=C:\Program Files\Java\jdk1.8.0_202\bin

C:\Users\Asmitha kadiri\Downloads>javac -version
javac 1.8.0_202

C:\Users\Asmitha kadiri\Downloads>javac TASK1.java

C:\Users\Asmitha kadiri\Downloads>java TASK1

C:\Users\Asmitha kadiri\Downloads>javac TASK2.java

C:\Users\Asmitha kadiri\Downloads>java TASK2
```

Online Food Ordering and Billing System Window:

The application window features a form with the following fields:

- Food Category: Main Course (dropdown)
- Food Item: Burger - 120 (dropdown)
- Quantity: 1 (text input)
- Add Item button

Below the form, the "CURRENT ORDER" section displays the following items and prices:

Item	Price
Brownie x 5	= Rs. 500.00
Farmhouse Pizza x 1	= Rs. 200.00
Fresh Juice x 2	= Rs. 160.00
Burger x 3	= Rs. 360.00
Subtotal:	Rs. 1300.00

At the bottom of the window, there are "Previous" and "Next" buttons.

JAVA PROGRAMMING LABORATORY

Evaluation Rubric

Criterion	Weightage	Marks / Remarks
Problem Understanding	15%	
GUI Components and Layout	20%	
Event Handling	20%	
Functional Correctness	25%	
Input Validation and User Experience	10%	
Program Explanation / Viva	10%	

Student Instructions

1. The solution must be implemented in Java using the concepts covered in the laboratory syllabus.
2. Students should demonstrate compilation, execution, and output during the practical session.
3. Use meaningful variable names and appropriate comments.
4. Students must be able to explain the selected AWT components, layout manager, event handling mechanism, and processing logic.